

EduTech Solutions

Internship Completion Report

Intern Details

Intern Name: Priya Dattatray Deshmukh

Start Date: April 20, 2026

Internship Duration: 2 Months

Intern ID: EDU-2026-870

Internship Domain: Data Science Intern

1. Project Title

Cryptocurrency Price Prediction using Deep Learning (LSTM)

2. Project Objective

The objective of this project is to develop a Cryptocurrency Price Prediction system using a Long Short-Term Memory (LSTM) neural network. The system analyzes historical Bitcoin price data and predicts future price trends based on previous market behavior. Historical data is collected through financial APIs, preprocessed using normalization techniques, and used to train the deep learning model. The

prediction system helps investors, traders, and researchers understand possible future market movements through data-driven forecasting.

3. Project Description in Detail

The Cryptocurrency Price Prediction System is a deep learning application designed to forecast Bitcoin prices using historical market data. The project utilizes an LSTM (Long Short-Term Memory) neural network, which is highly effective for sequential and time-series data because it can learn long-term dependencies between previous price movements.

Historical Bitcoin prices are downloaded using the Yahoo Finance API and stored in a structured dataset. The preprocessing phase includes cleaning the dataset, selecting the closing price, normalizing values using MinMaxScaler, and preparing training samples using the Sliding Window technique. The LSTM model is then trained to identify hidden temporal patterns in the data and predict future closing prices.

After training, the model is evaluated by comparing predicted prices with actual historical prices. The performance is measured using Root Mean Squared Error (RMSE). The trained model is saved for future predictions, and a Streamlit-based web application is developed to display real-time Bitcoin prices and predicted values through an interactive user interface.

This project demonstrates the practical implementation of Deep Learning, Time-Series Forecasting, Financial Data Analysis, API Integration, Data Visualization, and Model Deployment using Python.

4. Timeline Overview

Task	Activities Planned		Activities Completed
Task 1	Understanding Cryptocurrency Dataset		Studied Bitcoin historical price data and identified required features.
Task 2	Data Collection		Downloaded Bitcoin historical data using Yahoo Finance API.
Task 3	Data Preprocessing		Cleaned the dataset, removed missing values, and normalized data using MinMaxScaler.
Task 4	Exploratory Data Analysis		Visualized Bitcoin price trends and analyzed market behavior.
Task 5	Sliding Window Preparation		Prepared sequential training data using a 60-day sliding window.
Task 6	LSTM Model Development		Designed and implemented the LSTM neural network architecture using TensorFlow and Keras.
Task 7	Model Training		Trained the deep learning model using historical Bitcoin prices.
Task 8	Model Evaluation		Evaluated prediction performance using RMSE and prediction graphs.
Task 9	Prediction Module		Developed a script to predict future Bitcoin prices.
Task 10	Streamlit Dashboard		Built a web-based dashboard to display predictions and historical price charts.

Task 11	Model Testing	Tested the prediction accuracy with unseen data.
Task 12	Documentation	Prepared project documentation, screenshots, and presentation.
Project Start	Project Planning	Defined project objectives and implementation strategy.
Project End	Final Submission	Submitted source code, report, presentation, and demonstration successfully.

5. Project Development Methodology

The Cryptocurrency Price Prediction project followed a structured development methodology consisting of planning, data collection, preprocessing, model development, evaluation, deployment, and testing.

Initially, historical Bitcoin price data was collected through the Yahoo Finance API. The dataset was cleaned, normalized, and transformed into sequential training samples using the Sliding Window approach. An LSTM neural network was implemented using TensorFlow and Keras to learn historical price patterns. The model was trained over multiple epochs and evaluated using Root Mean Squared Error (RMSE). Finally, a Streamlit-based web application was developed to provide real-time price visualization and future price prediction through a user-friendly interface. Continuous testing and performance evaluation were performed throughout the project to improve prediction accuracy.

6. Output Screenshots

- Bitcoin Historical Price Graph
- LSTM Model Training Output
- Actual vs Predicted Price Graph
- Streamlit Dashboard Home Page
- Prediction Result Screen
- Model Evaluation (RMSE)
- Cryptocurrency Price Prediction Chart

7. Challenges Faced

Challenge 1: Data Collection and Quality

Historical cryptocurrency data obtained from APIs contained missing values, inconsistencies, and fluctuations due to market volatility. Ensuring the dataset was clean and reliable was essential before training the model.

Resolution:

Data preprocessing techniques such as handling missing values, removing invalid records, and normalizing the data using MinMaxScaler were applied to improve data quality.

Challenge 2: Time-Series Data Preparation

Unlike traditional machine learning models, LSTM requires sequential input data. Preparing the dataset using an appropriate sliding window size was challenging.

Resolution:

A 60-day sliding window technique was implemented to convert historical price data into input-output sequences suitable for LSTM training.

Challenge 3: Model Accuracy and Overfitting

Training the LSTM model without overfitting while maintaining good prediction accuracy required careful tuning of model parameters.

Resolution:

Dropout layers were introduced to reduce overfitting, and the model was trained over multiple epochs with the Adam optimizer and Mean Squared Error (MSE) loss function.

Challenge 4: Real-Time Data Integration

Integrating real-time Bitcoin price data with the prediction model required proper API handling and data formatting.

Resolution:

The Yahoo Finance API was used to fetch updated Bitcoin prices, and the same preprocessing pipeline was applied before generating predictions.

8. Learnings & Skills Acquired

Technical Skills

- Python Programming
- Deep Learning using TensorFlow and Keras
- Long Short-Term Memory (LSTM) Networks
- Time-Series Forecasting
- Data Preprocessing using Pandas and NumPy
- Data Normalization using MinMaxScaler
- Data Visualization using Matplotlib
- Real-Time API Integration using Yahoo Finance
- Model Evaluation using RMSE
- Streamlit Web Application Development

Soft Skills

- Analytical Thinking and Problem Solving
 - Time Management
 - Research and Self-Learning
 - Technical Documentation
 - Communication and Presentation Skills
 - Team Collaboration
 - Debugging and Error Resolution
-

9. Testimonial from Team

The Cryptocurrency Price Prediction project provided valuable hands-on experience in developing deep learning models for financial forecasting. Throughout the internship, the project enhanced practical knowledge of time-series analysis, data preprocessing, neural network implementation, and model evaluation. Working with real-world cryptocurrency data improved understanding of market trends and predictive analytics. Continuous testing, debugging, and collaboration contributed significantly to the successful completion of the project and strengthened both technical and professional skills.

10. Conclusion

The Cryptocurrency Price Prediction System was successfully developed using a Long Short-Term Memory (LSTM) deep learning model. The project demonstrated the complete workflow of a real-world AI application, including data collection, preprocessing, feature scaling, model training, evaluation, and deployment.

The trained model successfully learned historical Bitcoin price patterns and generated future price predictions based on previous market trends. A Streamlit-based web application was also developed to provide an interactive interface for displaying historical data and predicted cryptocurrency prices.

This project provided practical exposure to Deep Learning, Time-Series Forecasting, API Integration, Data Visualization, and Model Deployment. It also strengthened programming skills in Python, TensorFlow, Keras, Pandas, and Streamlit while demonstrating how Artificial Intelligence can be applied to solve real-world financial forecasting problems.

11. Acknowledgements

I would like to express my sincere gratitude to my mentor, **Geetanjali**, for her continuous guidance, encouragement, and valuable suggestions throughout the internship. Her technical support and constructive feedback greatly contributed to the successful completion of this project.

I also thank **EduTech Solutions** for providing me with the opportunity to work on a practical Artificial Intelligence and Machine Learning project that enhanced my technical knowledge and problem-solving abilities.

I am grateful to my institution and faculty members for their constant encouragement and support during the internship period. Special thanks to the developers and contributors of open-source technologies such as Python, TensorFlow, Keras, Streamlit, Pandas, NumPy, Matplotlib, Scikit-learn, and Yahoo Finance API for providing excellent resources that made this project possible.

Finally, I thank everyone who directly or indirectly supported me throughout this internship and contributed to the successful completion of the Cryptocurrency Price Prediction using Deep Learning (LSTM) project.

